trillion. China had roughly seven times India's total reserves, and the gap was $2.759 trillion.

In the long run, a key element of sustained economic growth is technological innovation. Measuring innovation is a complex matter, but one crude index is the number of scientific–technological patents that a country's nationals file. According to WIPO's Patent Cooperation Treaty System (PCT), from 2014 to 2019 India filed 10,012 patents in total, for an average of 1668 per year. In the same period, China filed 259,716 patents in total, for an average of 43,286 per year. Put differently, Chinese inventors filed roughly twenty-five times the number of patents that their Indian counterparts filed.[3]

Another measure of economic vitality is the number of market leaders that a country has – in essence, Fortune 500 companies. The US and China dominate, with China close to surpassing the Americans in terms of the total number. In 2019, India had seven out of 500 whereas China had 112 (the US had 121).[4] Clearly, India is nowhere in the league tables of the world's biggest companies. Nor can India match China in terms of the super-rich. In 2020, Mukesh Ambani was the richest person in Asia, but in the list of the 50 richest people in the world, India has only Ambani while China has nine magnates.[5] If we are looking at market leadership, in terms of companies and individuals, India fares very poorly compared to China – indeed there is no real comparison.

Transportation and Energy

Economic growth and development depend on infrastructure – transportation and energy. How do the two countries fare in terms of access to railways, highways, civilian aviation, and shipping as well as energy?

The total number of rail miles in India and China is about the same. In 2019, India had 123,542 kilometres, and in 2018 China had 131,651 kilometres. The total road network in India in 2017 was 5.897 million kilometres and in China in 2019 it was 5.012 million kilometres. India's